

Pneumonia Detection Using Deep Learning

Overview

This project implements a deep learning-based system for detecting pneumonia from chest X-ray images. The model is trained on a labeled image dataset containing Normal and Pneumonia cases and aims to assist in the early diagnosis of lung infections.

Features

- Chest X-ray image classification
- Automatic pneumonia detection
- Image preprocessing and augmentation
- Model training and validation
- Performance evaluation using standard metrics

Dataset

The dataset consists of chest X-ray images categorized into Normal and Pneumonia classes.

Technologies Used

Python, TensorFlow/Keras, NumPy, Pandas, Matplotlib, OpenCV, and Scikit-learn.

Workflow

Load images → Preprocess data → Apply augmentation → Train CNN → Validate model → Evaluate performance → Predict pneumonia from new X-rays.

Model Architecture

Convolutional Neural Network (CNN) with convolution layers, pooling layers, dropout layers, dense layers, and a classification output layer.

Performance Metrics

Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.